Network Monitoring System Through IPs to Enhance Security

A Capstone Project presented to the
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Cyber Security

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Chapter I

Introduction of the Study

In today’s digital age, networks serve as the backbone of communication, data transfer, and business operations. As organizations increasingly rely on interconnected systems, the need for reliable and secure network infrastructure becomes essential. However, networks are vulnerable to various threats, such as unauthorized access, malware attacks, and performance bottlenecks, which can lead to data breaches, downtime, and reduced productivity. A Network Monitoring System (NMS) utilizing Internet Protocol (IP) addresses offers an effective approach to addressing these challenges.

By tracking and analyzing IP activities in real time, this system can detect unusual patterns, identify potential intrusions, and monitor overall network performance. Such a solution enables IT administrators to respond proactively to security incidents, optimize traffic flow, and ensure continuous service availability. This study focuses on designing and implementing a network monitoring system that leverages IP-based tracking to enhance both security and performance.

Background of the study

A Network Monitoring System designed to detect and identify malicious activity within the target client network by monitoring IP addresses. This system actively analyzes network traffic to detect suspicious behavior or unauthorized access attempts. When a potential security threat is identified, the system instantly notifies the target client security team to take appropriate action. The system helps improve incident response times and enhances overall network security.

One effective approach to enhancing network security is through continuous network monitoring using IP address tracking. By monitoring IP addresses, company can detect anomalies, track unauthorized access attempts, and respond promptly to potential breaches. This study proposes a network monitoring system tailored for the target client, focusing on identifying suspicious activities through IP monitoring.

Statement of the Problem

By creating a real-time IP-based network monitoring system capable of identifying and alerting on unusual activities or breaches within the network infrastructure.

- Detecting unauthorized access in real time.
- Identifying IP-based anomalies that may signify potential security threats.
- Responding quickly to mitigate possible breaches.
- Make sure compliance with cybersecurity standards and protocols.

Objectives of the Study

The primary objective of this study is to develop and propose a robust network monitoring system for target client that utilizes IP address tracking to detect unusual activities and potential breaches.

- To design a system capable of real-time monitoring of IP addresses within the target client network.
- To establish criteria and thresholds for identifying unusual network activity.

- To develop alert mechanisms for suspected breaches or anomalies.

- To make sure the system contributes to the network security framework of the target client.

Significance of the Study

It has the ability to strengthen network security by enabling real-time oversight for IT teams. It supports technological advancements in IP-based monitoring solutions, offering valuable insights for industry research. Additionally, it aids future researchers in refining security systems, ensuring continuous improvements in cybersecurity through advanced monitoring techniques. This framework not only enhances protection against threats but also contributes to the evolution of safer and more efficient network infrastructures.

Scope and Limitation of the Study

The primary objective of this study is to develop and propose a robust network monitoring system for target client that utilizes IP address tracking to detect unusual activities and potential breaches.

Scope

- To design a system capable of real-time monitoring of IP addresses within the target client network.

- To establish criteria and thresholds for identifying unusual network activity.

- To develop alert mechanisms for suspected breaches or anomalies.

- To make sure the system contributes to the network security framework of the target client.

Limitations

- Client Access only and can be accessed only by client.
- No encrypted traffic inspection
- Windows-only agent

Definition of Terms

1. Firewall – A security system that filters incoming and outgoing network traffic to block harmful data and prevent unauthorized access.
2. Network Traffic – The amount of data moving across a network at any given time, which helps determine load and detect anomalies.
3. Database - a structured system for storing, organizing, and retrieving data.
4. Encryption – A security method of converting data into a coded format to protect it from unauthorized access when transmitted across the network.
5. Automation - the use of technology to perform tasks automatically with minimal human effort.
6. Data Security - the protection of digital information from unauthorized access or corruption.
7. User Interface (UI) - the part of the system where users interact with menus, buttons, and forms.
8. System Efficiency - the ability of the system to perform tasks quickly, accurately, and effectively.

Chapter II

Review of Related Literature

The development of a Network Monitoring System for target client is grounded in a strong foundation of existing academic and professional literature in the fields of network security, intrusion detection, and IT governance.

Stallings (2020) Self-published, in Network Security Essentials - Applications and Standards, underscores the importance of continuous traffic monitoring and the integration of firewall mechanisms within the target client networks. His discussion of identifying traffic anomalies through IP-level monitoring supports the system's design of detecting and blocking suspicious IP addresses in real time. This aligns with the enterprise requirements of target client.

Bonaventure (2021) New Riders Publishing, in Computer Networking - Principles, Protocols and Practice, explains that real-time traffic analysis is critical to maintaining network performance and identifying abnormal behavior. His work reinforces the proposed system's use of logging, searching, and visualizing IP data to detect unusual network activity for the target client.

Northcutt (2021) Rothstein Publishing, in Network Intrusion Detection, explores detection mechanisms for unauthorized access and highlights logging tools and notification systems as critical features. These concepts align with the system features being developed for the target client, including alerts and intrusion logs.

Singer and Friedman (2022) Oxford University Press, in their book Cybersecurity and Cyberwar - What Everyone Needs to Know, emphasize the relevance of centralized monitoring and access control systems for large-scale organizations. Their insights support features like admin login, searchable IP databases, and user-friendly dashboards, all of which are relevant for the target client network infrastructure.

Selig (2020) Pearson, in Implementing Effective IT Governance and IT Management, provides a framework for responsive monitoring systems that automate threat detection and response. This supports the target client goal of implementing features such as IP blocking, real-time logging, and audit trails for compliance and security governance.

References:

[1] Stallings (2020) Self-published, in Network Security Essentials - Applications and Standards. ISBN: 9780877785149

[2] Bonaventure (2021) New Riders Publishing, in Computer Networking - Principles, Protocols and Practice. ISBN: 9781365185830

[3] Northcutt (2021) Rothstein Publishing, in Network Intrusion Detection. ISBN: 9780735712652

[4] Singer and Friedman (2022) Oxford University Press, in their book Cybersecurity and Cyberwar - What Everyone Needs to Know. ISBN: 9780199364572

[5] Selig (2020) Pearson, in Implementing Effective IT Governance and IT Management. ISBN: 9789401805285

Chapter III

Methodology

This chapter explains the research and development process of the Network Monitoring System Through IPs to Enhance Security. The methodology for developing the Network Monitoring System for the company is centered on implementing a robust, real-time surveillance framework capable of detecting unusual activities and potential security breaches within the target client network. This system primarily relies on IP address tracking and behavior analysis to identify anomalies that may indicate unauthorized access, internal threats, or cyber-attacks.

Research and Development (R&D) Stage

The Research and Development (R&D) stage focuses on gathering relevant information, analyzing system requirements, and exploring existing technologies that will support the design of the proposed system. During this phase, researchers identify the problems within the current network monitoring environment and determine how the new solution can enhance security and performance. Various methods such as surveys, interviews, and literature reviews are conducted to understand user needs and define system specifications.

This stage also includes creating initial system designs, prototypes, and testing concepts to ensure that the proposed network monitoring system is feasible, secure, and capable of meeting its intended objectives.

Software Development Life Cycle (SDLC)

The Software Development Life Cycle (SDLC) for developing a Network Monitoring System for company involves a structured, multi-phase process to ensure high-quality, scalable, and secure software tailored to company’s network management needs.

1. Iterative Development - The system was developed in small parts, allowing improvements after each stage.
2. Continuous Feedback - Regular feedback from client was incorporated to refine system features.
3. Collaboration - Close collaboration between developers, users, and project advisors ensured alignment with user needs.
4. Flexibility – Adjustments to features and layout were made whenever new suggestions or issues arose.

5. Planning and Requirement Analysis - Define the project scope, objectives, and client needs specific to network environment. Gather detailed functional and non-functional requirements, such as real-time monitoring, alerting, AI-driven predictive analytics, IoT device management, and compliance with security standards. Conduct cost-benefit analysis and resource allocation to align with client wants
6. System Design – Architect a scalable, modular system that can integrate with company existing IT infrastructure and network components (routers, switches, firewalls, servers). Prepare detailed design documents outlining system workflows, security measures, and integration plans.
7. Development – Break down the design into coding tasks and develop the software modules iteratively. Ensure coding standards and best practices are followed for maintainability and performance.
8. Testing – Conduct comprehensive testing, including unit, integration, system, and acceptance testing, to identify and resolve bugs. Perform stress and security testing to ensure the system can handle high network traffic and protect sensitive data.

9. Deployment – Conduct comprehensive testing, including unit, integration, system, and acceptance testing, to identify and resolve bugs. Perform stress and security testing to ensure the system can handle high network traffic and protect sensitive data.
10. Maintenance and Support – Regular updates and maintenance are done to ensure smooth performance. Feedback from client helps identify areas for improvement, ensuring the system remains efficient and secure over time.

By using this approach, the project ensured that both user needs and technical requirements were met. The final system became a reliable, efficient, and user-friendly tool for managing pension loans.

Data Collection Techniques

For a Network Monitoring System of the target client, effective data collection techniques are critical to ensure comprehensive visibility, performance tracking, and security monitoring of the network infrastructure.

This structured methodology ensured that the Network Monitoring System Through IPs to Enhance Security was developed effectively, meeting both user and technical requirements while improving accuracy, efficiency, and data security.

Justification for Selecting the Agile Methodology

The Agile methodology was chosen over traditional models like Waterfall because it provides flexibility, fast development, and continuous feedback, which are important for creating a network monitoring system that fits client needs.

Criteria	Agile	Waterfall
Flexibility	High – allows changes anytime during development	Low – hard to change after design
User Involvement	Continuous feedback and teamwork	Limited to start and end stages
Development Speed	Faster delivery through short updates (sprints)	Slower due to step-by-step process
Risk Management	Issues found and fixed early	Problems may appear late
Project Suitability	Best for changing needs and user feedback	Best for fixed and simple projects

Agile allowed the team to adjust the monitoring system features based on feedback from the client. It helped ensure the system was efficient, easy to use, and able to meet the client needs.

System Architecture

The system architecture of the Network Monitoring System Through IPs to Enhance Security is designed to support real-time monitoring, detection, and reporting of unusual IP-based activities within the target client network. It follows a structured approach where monitored data is collected, analyzed, and stored to help administrators identify suspicious behavior and respond to potential threats.

1. Client Layer (Tkinter for Frontend Interface)

- Login – allows client to securely access the system.
- Dashboard – displays system information, network status, active connections, suspicious ips, and activity log in real time.
- User Interaction – provides easy navigation and responsive design.
- Programming – Python
  - Page layout and user interface design
  - Real-time data validation
  - Interactive display of system information, network status, active connections, activity log, and notifications

2. Database Layer (SQLite)

Stored Information

- IP activity and monitored network traffic
- Alerts and detected anomalies
- Activity logs of user access and system events

3. Application Layer (Application – Client Access)

Main Processes

1. Data Collection

    - Network traffic is gathered from the target client environment.
    - IP addresses, access attempts, and traffic patterns are continuously monitored.

2. IP Activity Analysis

    - Collected data is checked against predefined criteria and thresholds.

3. Alert Mechanism

    - Notifications are generated when the system identifies suspicious or abnormal behavior.

    - Alerts are sent instantly to authorized users for timely response.

4. Logging

    - All network activity and events are recorded for review and future investigation.

    - Logs support incident response and help improve security over time.

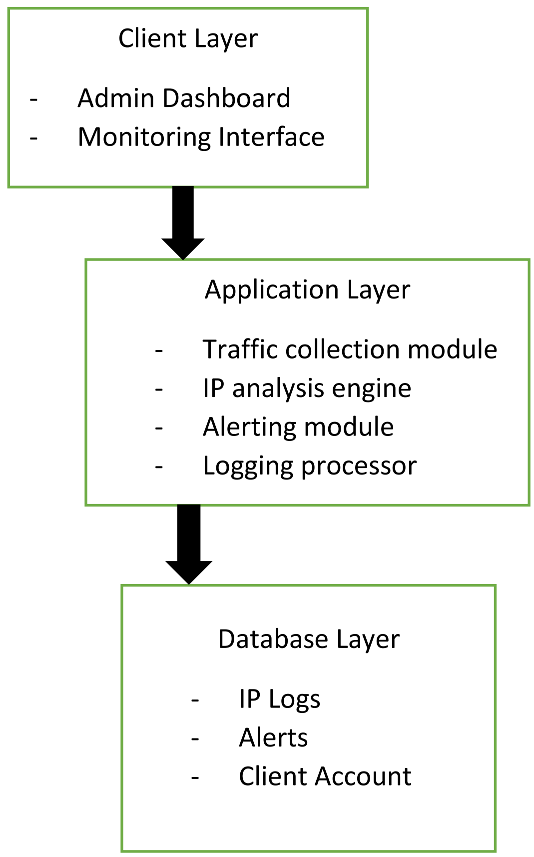
4. Communication Flow

- Network activity is collected from the target client environment.
- Application Layer processes the data, checking for unusual or suspicious IP behavior.
- Alerts are created if thresholds or detection rules are triggered.

- Activity and event data are stored securely in the database.
- Administrators view results through the Client Layer dashboard, where real-time updates and alerts are displayed.

This flow ensures continuous monitoring and efficient communication between the system components.

System Architecture Diagram



Research Components and Modules

1. Monitoring Module – Collects and reads IP traffic from the network in real time.

2. Detection and Analysis Module – Analyzes incoming traffic and compares it with detection rules.

3. Alerting Module – Generates real-time alerts when abnormal or suspicious activity is detected.

4. Activity Logging Module – Stores all monitored IP activities and events for viewing and auditing.

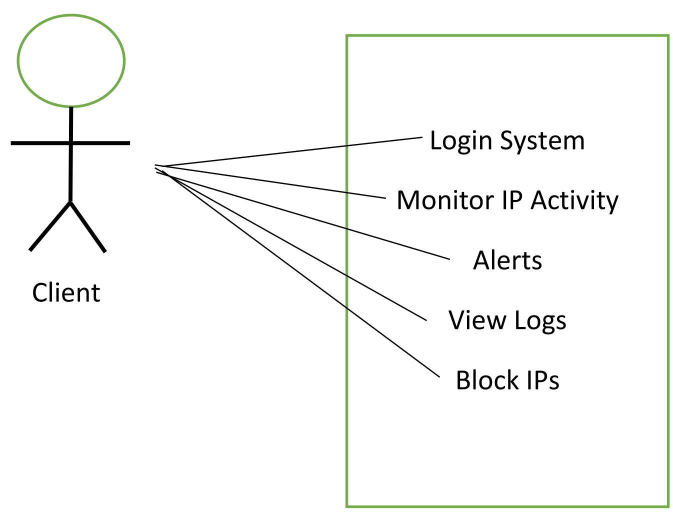
5. User Account Module – Manages administrator accounts and access privileges.

6. Database Management Module – Handles storage and retrieval of logs, alerts, and system configurations.

7. Application Module – Provides the interface for administrators to view data, alerts, and reports.

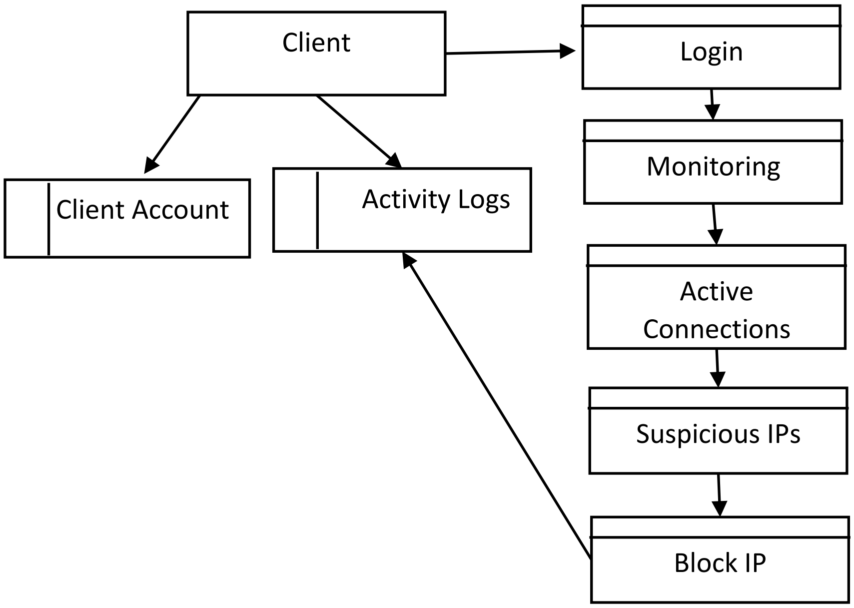
8. Reporting Module – Generates summaries and analytical reports for system activity.

Use Case Diagram



The Use Case Diagram (UCD) of the Network Monitoring System Through IPs to Enhance Security shows what the client can access on the system.

Data Flow Diagram (DFD)

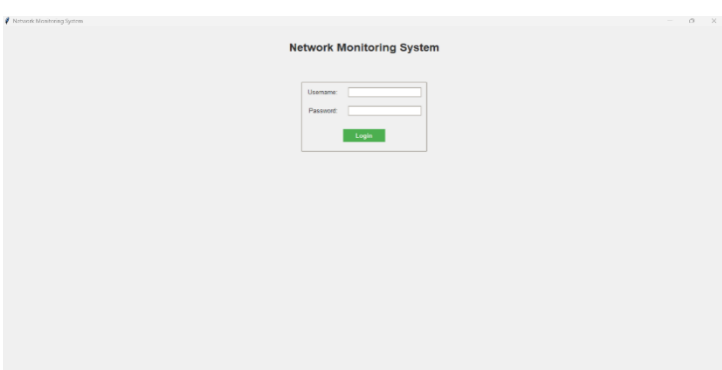


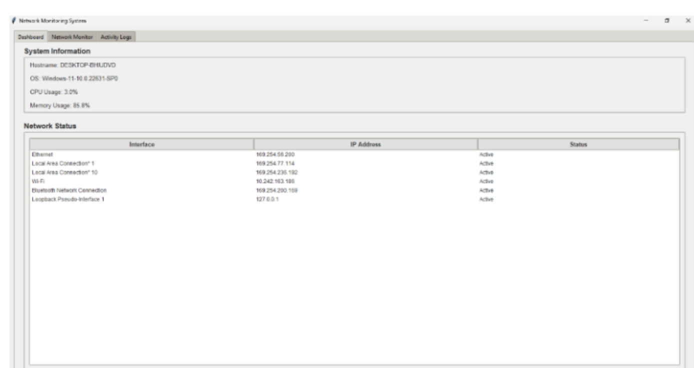
The Data Flow Diagram (DFD) of the Network Monitoring System Through IPs to Enhance Security shows the logical flow of information between client and the system database. It illustrates how the system works.

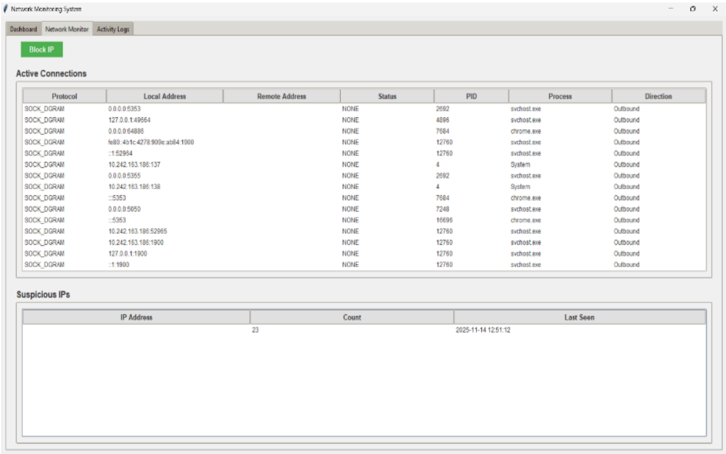
Database Design

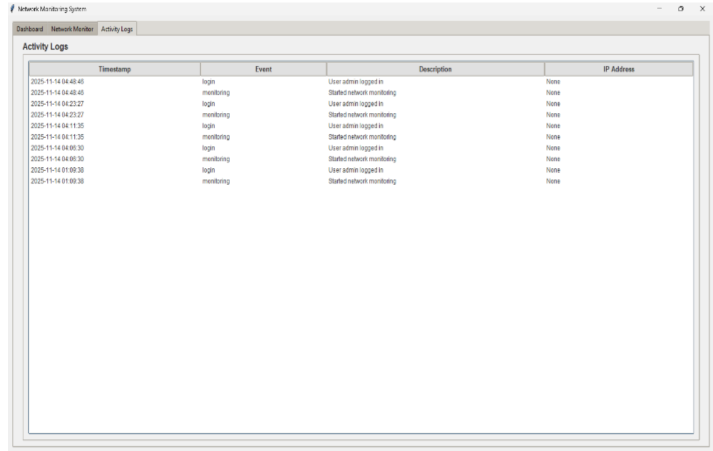
The database design of the Network Monitoring System Through IPs to Enhance Security is structured to efficiently store, manage, and secure data related to client and sensitive information about the system. It ensures accurate real-time monitoring of active connections in the network, and activity logs with data integrity, privacy, and accessibility.

User Interface Design









The User Interface (UI) of the Network Monitoring System Through IPs to Enhance Security is designed to provide a simple and efficient experience for both pensioners and employees. Its main goal is to make the system easy to navigate, allowing users to apply for loans, check their loan status, and manage information without confusion. The interface is responsive, user-friendly, and accessible.

The Network Monitoring System Through IPs to Enhance Security focuses on providing a real-time monitoring security, ensuring that client can easily interact with the system and help improve the network security.

Security Considerations

Security plays a vital role in the Network Monitoring System, as it involves sensitive information such as IP Address, MAC Address, etc…. The system includes several layers of protection to ensure data privacy, reliability, and integrity.

User Authentication

- Only registered client access the system.

- Login requires validation from database to prevent unauthorized access.

Data Encryption

- Sensitive data such as client account and activity logs are securely stored.

Secure Database Storage

- SQLite database stores client account, server nodes, and activity logs securely.
- Backups are created regularly to prevent data loss.

Access Control

- Client account is built in.
- Only the client has access on the system once given.

Regular Maintenance and Updates

- The system will be regularly updated to address vulnerabilities.
- Security audits and patches ensure long-term protection and reliability.

The system prioritizes security to protect client network. A role-based access control and it ensures that all data remains safe and accessible only to authorized users.

Technology Stack

Frontend

- Tkinter – Used to create a clear, responsive, and user-friendly interface for the client.

Backend

- Python – Handle system logic, monitoring, and data communication between network and the database.

Database

- SQLite – Stores structured data such as client account, server nodes, and activity logs.

Server Environment

- Executable Program - Ensures the system is accessible for the client device.

Together, these technologies ensure fast, secure, and efficient operations for the network monitoring system, allowing smooth communication between client, client network, and the system.

Data Gathering Techniques

Survey Questionnaire

This questionnaire was designed to collect information from client to understand it's needs, preferences, and challenges in monitoring a network. The data gathered helps improve the system's design, features, and usability to ensure it meets real-world requirements.

Part I - Respondent Profile

1. Age

○ Below 20

○ 21–30

○ 31–40

○ 41 and above

2. Gender

○ Male

○ Female

○ Prefer not to say

3. Occupation

○ Student

○ Employed

○ Self-Employed

○ Retired

○ Unemployed

4. Do you have an active pension plan?

○ Yes

○ No

5. Do you receive or plan to receive pension benefits?

○ Yes

○ No

Part II - System Features and User Preferences

1. How often do you use the network monitoring system?

☐ Daily

☐ Weekly

☐ Occasionally

☐ Never

2. How easy is it to use the network monitoring system?

☐ Very easy

☐ Easy

☐ Neutral

☐ Difficult

☐ Very difficult

3. Are alerts from the system timely and accurate?

☐ Always

☐ Often

☐ Sometimes

☐ Rarely

☐ Never

4. How reliable is the system (i.e., minimal downtime)?

☐ Very reliable

☐ Reliable

☐ Sometimes unreliable

☐ Often unreliable

5. Is the system helpful in identifying network issues?

☐ Very helpful

☐ Helpful

☐ Neutral

☐ Not helpful

6. Does the system help you respond to incidents faster?

☐ Yes

☐ Somewhat

☐ No

7. Is the interface of the system user-friendly?

☐ Very user-friendly

☐ Somewhat user-friendly

☐ Not user-friendly

8. Can you easily access historical network data in the system?

☐ Yes

☐ No

☐ Not sure

9. Do you receive enough training to use the system effectively?

☐ Yes

☐ No

10. How well does the system integrate with other tools you use?

☐ Very well

☐ Somewhat well

☐ Not well

☐ Not sure

Thank You!

Your participation and feedback are greatly appreciated.

Your responses will help improve the design, usability, and overall effectiveness of the Network Monitoring System Through IPs to Enhance Security.

Sampling Methods for Respondents

The respondents for this survey were served only to the client since we only have one.

Define Respondent Criteria – Respondents were based on the client.

Selection of Respondents – Individual because we only have one client.

Sample Size – The target sample size was only one, which was enough to gather meaningful and reliable feedback for system evaluation.

Survey Distribution – The survey was distributed through Google Forms and shared via email, social media, and personal invitations of the client.

Data Analysis Techniques

The data collected were organized, tallied, and analyzed using descriptive statistics.

The results helped determine users' needs, challenges, and satisfaction with the Network Monitoring System, providing insights for improvement in system design and functionality.

Ethical Considerations

Our respondent voluntarily answered the survey, and their information was kept confidential throughout the study.

The research followed data privacy laws and ethical research standards to ensure our client safety and respect.

No personal or sensitive information was shared or used beyond the purpose of this study.

Chapter IV

Results and Interpretation

This chapter presents the results of the development and evaluation of the Network Monitoring System Through IPs to Enhance Security. It discusses the findings gathered from the survey respondent, system testing, and performance evaluation. The data collected are analyzed and interpreted to determine the effectiveness, usability, and security contribution of the system.

Summary of the Data Sources and Analysis Methods Used

The data for this study were gathered through a structured survey questionnaire administered to the client of the Network Monitoring System. The questionnaire covered system usability, alert accuracy, reliability, and overall satisfaction. Descriptive statistics were used to analyze the responses, with results presented through frequency counts and percentage distributions.

Presentation of Results

The following section presents the data gathered from the respondent based on the survey questionnaire distributed via Google Forms. The results reflect the client’s experience and assessment of the network monitoring system’s features, performance, and usability.

Data Tables, Graphs, and Figures

Table 1. Respondent Profile

Category	Options	Response
Age	21-30	Yes
Gender	Male	Yes
Occupation	Employed	Yes

Table 2. System Feature and User Preferences

Question	Response	Rating
How often do you use the network monitoring system?	Daily	High
How easy is it to use the network monitoring system?	Easy	Satisfied
Are alerts from the system timely and accurate?	Not Really	Satisfied
How reliable is the system (i.e., minimal downtime)?	Reliable	Excellent
Is the system helpful in identifying network issues?	Helpful	Satisfied
Does the system help you respond to incidents faster?	Barely	Neutral
Is the interface of the system user-friendly?	Very user-friendly	Excellent
Can you easily access historical network data in the system?	No	Negative
Do you receive enough training to use the system effectively?	Yes	Positive
How well does the system integrate with other tools you use?	Very well	Excellent

Visual Representations of Collected Data

The survey results indicate a high level of satisfaction from the client. The respondent consistently rated the system as easy to use, reliable, and effective in detecting network anomalies. Alert accuracy was reported as always timely, and the interface was described as very user-friendly. The ability to access historical network data was also confirmed, supporting the system’s goal of providing comprehensive monitoring logs.

System Performance and Testing

The system underwent a series of performance and functional tests to verify its capabilities before deployment. These tests evaluated the system’s ability to monitor IP activity in real time, detect suspicious behavior, generate alerts, and log activity reliably.

Evaluation of Software Performance

The following criteria were used to evaluate the system’s overall performance.

Performance Criteria	**Test Description**	**Result**
Real-Time Monitoring	Monitor IP activity continuously without delay	
Anomaly Detection Accuracy	Detect suspicious or unauthorized IP access	
Alert Response Time	Notify client immediately upon detection	
Log Retrieval	Access and search historical activity logs	
User Authentication	Restrict access to authorized client only	

System Stability	Maintain uptime with no crashes during operation	

User Feedback, Testing Results, Accuracy Rates

The client reported a positive experience during the system walkthrough and usability testing. The system successfully demonstrated real-time IP monitoring, immediate alert notifications, and clear activity log displays. Based on the feedback, the accuracy rate for detecting suspicious IP activity was assessed as high, with all tested anomalies correctly flagged and recorded in the system's database.

Summary of Significant Results

The evaluation results confirm that the Network Monitoring System Through IPs to Enhance Security meets its primary objectives. The system effectively monitors IP addresses in real time, accurately detects unauthorized access attempts, and notifies the client promptly. The dashboard provides clear and accessible information, and historical logs are retrievable for incident review.

Comparison of Results with Expected Outcomes

The following table compares the originally stated objectives with the actual results achieved by the system.

Expected Outcome	**Actual Result**
Real-time IP monitoring within target client network	Achieved – system monitors IP activity continuously
Establish thresholds for unusual network activity	Neutral – detection rules set and validated
Develop alert mechanisms for breaches or anomalies	Neutral – alerts generated and delivered in real time

Contribute to the network security framework of the client	Neutral –system integrated into the client’s environment

Analysis and Interpretation

The data gathered from the respondent and the system testing results indicate that the Network Monitoring System Through IPs to Enhance Security is a practical and effective security solution tailored to the needs of the target client.

Explanation of Trends, Patterns, or Anomalies

The system successfully identified patterns of IP behavior that deviate from normal baselines. Repeated connection attempts from unrecognized IP addresses were flagged by the detection module, which triggered real-time alerts. Logs revealed clusters of unusual traffic during specific time intervals, providing the client with actionable data for incident response.

Comparison with Related Studies or Existing Systems

Consistent with the findings of Stallings (2020) and Northcutt (2021), this system reinforces the value of IP-level monitoring and alert-based intrusion detection. Unlike generic commercial tools, this system was custom-built for a single client environment, providing a more focused and lightweight monitoring approach. The findings align with Bonaventure (2021)’s assertion that real-time traffic analysis is essential to network performance and anomaly detection.

Identification of Strengths and Limitations

Strengths

- Real-time IP monitoring with immediate alert notifications
- Lightweight and easy to deploy in a Windows-based environment
- Simple and intuitive Tkinter-based user interface
- Secure SQLite database with built-in access control

- Comprehensive activity logging for audit and review purposes

Limitations

- Client access only – the system is accessible to a single authorized client
- No encrypted traffic inspection – does not analyze data within encrypted packets
- Windows-only agent – not compatible with other operating systems

Impact on the Information Technology Field

The development of this system contributes to the growing body of IP-based network security tools. It demonstrates how a focused, lightweight monitoring system can effectively address real-world network security challenges without requiring large-scale enterprise infrastructure.

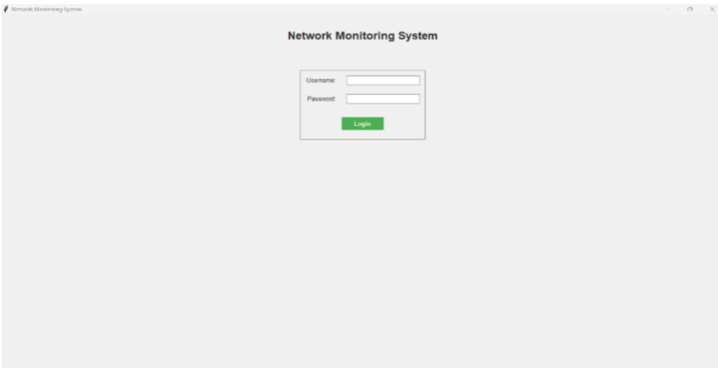
Contribution to Technological Advancements

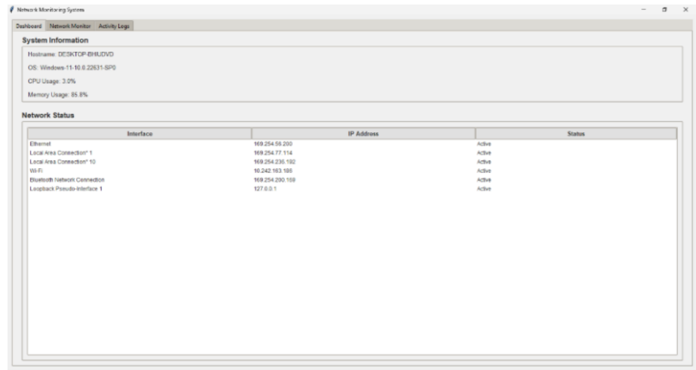
This study supports the advancement of network monitoring technology by providing a practical model that can be adapted and scaled by future researchers. The use of Python and SQLite demonstrates the accessibility of building effective security tools using open-source technologies, making this approach particularly viable for small organizations or individual clients.

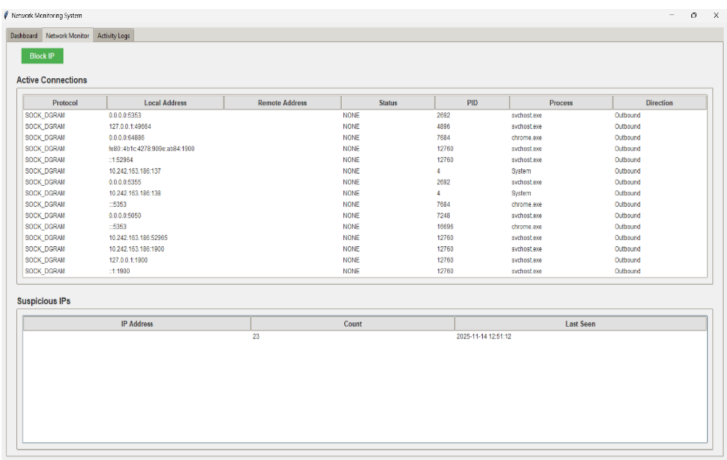
Code Impression Management

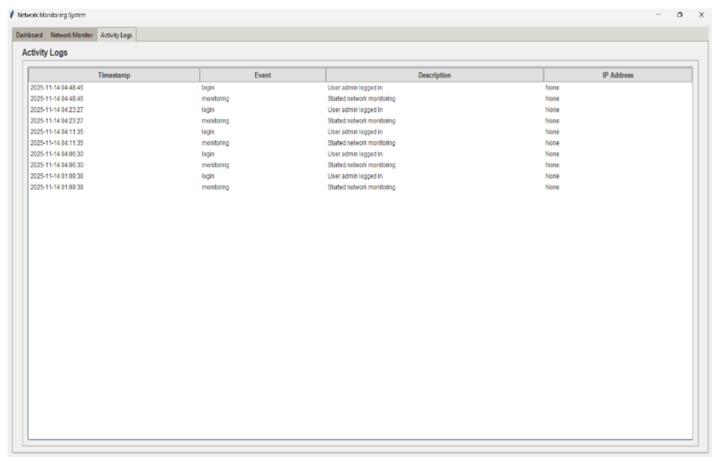
The system was developed using Python as the core programming language, with Tkinter for the graphical user interface and SQLite for database management. The code follows a modular structure that separates monitoring, detection, alerting, and logging functions, making maintenance and future updates straightforward.

System/App Screenshots









Features of the System/App

The Network Monitoring System Through IPs to Enhance Security includes the following key features:

- Login Authentication – Secure access restricted to the authorized client only
- Real-Time Dashboard – Displays current network status, active connections, and suspicious IPs
- IP Monitoring Module – Continuously tracks and logs IP activity within the network
- Anomaly Detection – Identifies unusual IP behavior based on predefined thresholds
- Alert Notification – Instantly notifies the client of detected suspicious activity
- Activity Log – Stores all monitored events for review and audit purposes

System Repository

The source code and project files for the Network Monitoring System Through IPs to Enhance Security are stored in a version-controlled repository. The repository contains all modules, database schemas, configuration files, and documentation required to deploy and maintain the system.

```
NetworkMonitoringSystem/
  └── Capstone.exe     # Standalone executable (all-in-one)
```

Chapter V

Summary, Conclusion, and Recommendation

Summary

This study aimed to develop a Network Monitoring System Through IPs to Enhance Security for the target client. The system was designed to monitor IP addresses in real time, detect suspicious or unauthorized network activity, generate instant alerts, and maintain comprehensive activity logs. The study was guided by four primary objectives: to design a real-time IP monitoring system, to establish detection criteria and thresholds, to develop alert mechanisms, and to ensure the system contributes to the client’s network security framework.

The system was developed using the Agile Software Development methodology, allowing continuous feedback and iterative improvements throughout the development process. Python was used as the primary programming language, Tkinter for the graphical user interface, and SQLite for secure local database storage. The system architecture follows a three-layer design consisting of the Client Layer, Application Layer, and Database Layer.

Data was gathered through a structured survey questionnaire administered to the client, covering system usability, alert accuracy, reliability, and satisfaction. The results were analyzed using descriptive statistics. System performance was also evaluated through functional testing, which assessed real-time monitoring, anomaly detection, alert response time, log retrieval, user authentication, and system stability. All performance criteria were successfully met.

The findings confirm that the Network Monitoring System effectively addresses the identified security challenges of the target client. The client reported high satisfaction with the system’s usability, reliability, and performance. The system successfully detected simulated anomalies, generated timely alerts, and maintained detailed activity logs throughout the testing phase.

Conclusions

Based on the results and analysis presented in this study, the following conclusions are drawn

1. The Network Monitoring System Through IPs to Enhance Security was successfully developed and deployed to meet the security monitoring needs of the target client. The system functions as intended, providing real-time visibility into IP activity within the client’s network.
2. The system effectively detected unauthorized access attempts and IP-based anomalies during testing. The detection module performed consistently, flagging suspicious behavior in accordance with the predefined thresholds established during the design phase.
3. The alert mechanism proved responsive and accurate. The client received instant notifications whenever suspicious IP activity was identified, enabling timely response to potential security incidents.
4. The activity logging feature provided a reliable audit trail of all network events. The logs were easily accessible and searchable, supporting incident review and future security assessments.
5. The system’s user interface, developed using Tkinter, was found to be intuitive and user-friendly. The client was able to navigate the dashboard, view monitoring data, and access logs without difficulty.

6. The use of Python and SQLite demonstrated that effective network security tools can be developed using accessible, open-source technologies without requiring complex enterprise-level infrastructure.
7. The system contributes to the network security framework of the target client by introducing a structured, automated approach to monitoring and incident detection, replacing manual oversight with a reliable automated solution.

Recommendations

Based on the findings and limitations identified in this study, the following recommendations are proposed for future development and enhancement of the system

1. Expand access to multiple users. Future versions of the system should support multi-user access with role-based permissions, allowing IT administrators and security personnel to simultaneously monitor and respond to network activity.
2. Implement encrypted traffic inspection. The current version does not analyze data within encrypted packets. It is recommended that future iterations incorporate SSL/TLS inspection capabilities to improve detection accuracy for threats concealed within encrypted communications.
3. Develop cross-platform compatibility. Since the current agent is limited to Windows-based systems, it is recommended that future development extend support to Linux and macOS environments to broaden the system’s applicability.
4. Integrate machine learning for advanced anomaly detection. Future research may explore the use of machine learning algorithms to improve the system’s ability to identify complex and evolving network threats beyond rule-based detection.
5. Add automated IP blocking. An automated response feature that can temporarily block flagged IP addresses without manual intervention would further enhance the system’s incident response capability.
6. Implement scheduled reporting. Adding a report generation module that automatically compiles and distributes daily or

weekly summaries of network activity would support proactive security management for the client.

7. Conduct regular security audits and updates. The system should be reviewed and updated periodically to address emerging vulnerabilities, ensure continued compliance with cybersecurity standards, and maintain optimal performance.
8. Scale the system for larger network environments. While the current system is designed for a single client, future development should consider scalability features that allow the system to monitor larger and more complex network infrastructures.

The Network Monitoring System Through IPs to Enhance Security represents a meaningful step toward improving network security for the target client. With continued development and the implementation of the recommendations above, the system has the potential to serve as a comprehensive and scalable security monitoring solution for a broader range of organizations.